When the Questionnaire Speaks isiXhosa but the System Speaks English: Adapting the System Usability Scale in South Africa

Inhle Cele
University of Cape Town
Cape Town, South Africa
clxinh001@myuct.ac.za

Abstract

The System Usability Scale (SUS) is widely used to measure perceived usability, but its standard form is written in English. This study developed and evaluated an isiXhosa adaptation of the SUS with 28 isiXhosa first/home-language speakers in South Africa, each completing a written English condition first and a written isiXhosa condition second, on different systems. A system-adjusted evaluation-level model found no statistically detectable within-person difference between conditions (isiXhosa minus English: 0.97 SUS points, 95% CI [−4.38, 6.32], $p = .713$), consistent with simpler descriptive comparisons; internal consistency was good in both conditions ($\alpha = .843$ and .890). An exploratory system-adjusted language-by-comfort interaction also did not detect a subgroup difference (2.43 points, 95% CI [−8.21, 13.07], $p = .643$). Reflexive thematic analysis of qualitative feedback from 14 participants suggested that English often felt easier because of familiarity with written English and technology terminology rather than English being inherently clearer, and that the easiest language was not always the preferred one. Because English always preceded isiXhosa and participants evaluated different systems in each condition, these results do not establish measurement equivalence or an isolated language effect. The study contributes a documented isiXhosa adaptation process and highlights written-language familiarity and register as factors in multilingual usability evaluation.

CCS Concepts

• **Human-centered computing** → **Usability testing**; *Accessibility design and evaluation.*

Keywords

System Usability Scale, SUS, isiXhosa, multilingual usability, cross-cultural adaptation, questionnaire translation, South Africa

1 Introduction

The System Usability Scale (SUS) is a widely used ten-item questionnaire for measuring perceived usability across different systems and contexts [12]. Its standardised scoring and low administration burden make it useful for comparative usability studies. However, like many standardised instruments, SUS was developed in English, raising questions about how well it functions when respondents routinely use other languages.

This is particularly relevant in South Africa, which has twelve official languages and where many people use English as an additional rather than a home language. Historical and institutional factors have nevertheless made English prominent in education and other formal settings [28, 29]. Consequently, an isiXhosa first-language speaker may be highly familiar with written English while using isiXhosa more frequently in other contexts. A translated questionnaire may therefore differ from its English version not only in language, but also in participants' familiarity with its written register.

This study develops and evaluates an isiXhosa adaptation of the SUS among isiXhosa first/home-language speakers. It forms part of the broader SUSEVAL project, which also included oral English and written Afrikaans conditions; this paper reports only the isiXhosa component. The study is additionally relevant to UCT's multilingual context, where institutional language policy recognises English, isiXhosa and Afrikaans as official languages and promotes the use and development of marginalised South African indigenous languages in teaching, learning, research and communication [34]. The study addresses the following research questions:

RQ1: Among isiXhosa first/home-language speakers, how do SUS scores differ between the written English and written isiXhosa questionnaire conditions?

RQ2: How do English–isiXhosa SUS score differences vary according to participants' self-reported most comfortable language?

*H*ere, "most comfortable language" refers specifically to the language participants reported feeling most comfortable completing the study in, rather than to an objective measure of language proficiency.

The primary hypothesis was:

H1: SUS scores will differ between the written English and written isiXhosa conditions.

RQ2 is exploratory because most comfortable language was selected as a grouping variable after inspection of the collected demographic distributions, as described in Section 3.2, in place of an originally planned proficiency-band comparison. Educational attainment was also examined after the same inspection; because it was not part of the original analysis plan, it is reported separately as an exploratory appendix analysis (Appendix A.2) rather than as part of RQ2.

This work makes three contributions. First, it documents the development of an isiXhosa SUS through forward translation, synthesis, back-translation and pretesting. Second, it provides an initial empirical comparison of written English and isiXhosa SUS responses while explicitly accounting for differences in the systems evaluated. Third, it shows through participant accounts that written-language familiarity, register and language preference can shape the experience of a translated usability questionnaire even when average SUS scores are similar.

2 Background Work

2.1 SUS and Multilingual Use

SUS consists of ten alternating positively and negatively worded items rated on a five-point scale and produces a perceived-usability score from 0 to 100 [12]. Higher scores indicate more favourable perceived usability of the evaluated system; a higher score does not by itself indicate that one questionnaire translation is more accurate. It has been widely adopted because it is brief, system-independent and supported by substantial reliability evidence. Bangor et al., for example, reported high internal consistency (α = .91) across more than 2,300 administrations [2], while later reviews found strong evidence for its use as a standardised post-study usability measure [19].

SUS has also been translated into numerous languages. Published adaptations generally report acceptable internal consistency, although sample sizes, validation methods and translation procedures vary considerably [16, 21]: across six such translations (Arabic, Slovene, Polish, Italian, Persian, Portuguese), reported α ranges from .79 to .84, with validity evidence reported inconsistently or not at all (Appendix Table 5).

The reviewed studies do not apply a uniform set of psychometric assessments: some report internal consistency without a validity coefficient, whereas others report validity evidence without Cronbach's alpha. This study builds on that uneven evidence base but does not attempt a full validation.

2.2 Translation and Cross-Cultural Adaptation

Questionnaire translation requires more than lexical substitution. Beaton et al. distinguish semantic, idiomatic, experiential and conceptual equivalence, all of which may affect whether an adapted item measures the same construct [4]. Recommended procedures therefore combine forward translation, synthesis, independent back-translation, expert review and pretesting with the target population [4, 26].

Psychometric assessment can then examine properties such as internal consistency and test–retest reliability [6, 31, 32]. Importantly, high internal consistency does not by itself establish linguistic or conceptual equivalence. For this reason, translation development and quantitative performance are considered separately when evaluating the isiXhosa SUS.

2.3 South African Context and Research Gap

SUS has been used in South African studies of accessibility technologies, HIV self-testing, and online banking [18, 22, 25]. However, the reviewed studies do not evaluate an isiXhosa SUS or examine how questionnaire language affects responses. South African work outside HCI shows that structured isiXhosa questionnaire adaptation using forward translation, back-translation and review is feasible [3, 33].

The gap is not only the absence of an isiXhosa translation. It is also unclear how written isiXhosa SUS responses behave relative to written English, how internal consistency compares across conditions, and how participants experience the translated wording and register. To address this gap, the study combines a documented adaptation process, a within-participant English–isiXhosa comparison, and qualitative feedback.

3 Design and Experimental Methodology

3.1 Research Design and Research Questions

The isiXhosa component used a within-participant design with a fixed language sequence and manually varied system assignments. Every participant completed the written English condition first and the written isiXhosa condition second, rating three assigned systems in each condition using the ten-item SUS. SUS items alternate positive and negative wording on a five-point agreement scale and are combined using the standard odd/even scoring procedure into a 0–100 score per evaluation [12]. For each participant the researcher selected three systems per phase, aiming to keep system representation reasonably balanced across the sample and to avoid same-system repetition within a participant; assignments were varied pragmatically rather than generated by a formal schedule or randomisation, and the intended balance was not achieved in practice. Figure 1 summarises the procedure.

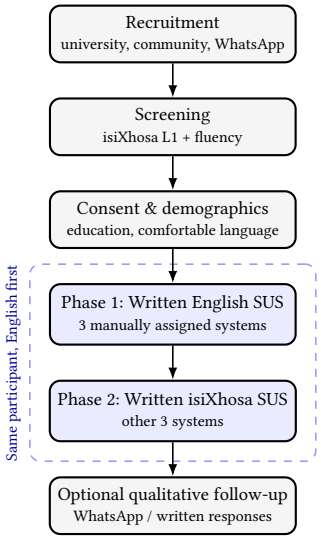


Figure 1: isiXhosa study procedure.

This study is a preliminary evaluation of the adaptation, focusing on score patterns, internal consistency and participant feedback. It does not attempt the full psychometric validation set out in Section 2.2, which would require a larger and more diverse sample together with test–retest reliability, convergent or criterion validity, confirmatory factor analysis and measurement invariance between language versions; those analyses are identified as future work (Section 5).

3.2 Participants and Recruitment

Participants were adults aged 18 years or older, recruited through university and community channels, including posters, WhatsApp, community networks, and the UCT Xhosa Society. Participants entered the isiXhosa component if they reported isiXhosa as their

first/home language and reported fluency in it; allocation was therefore purposive rather than random.

The final isiXhosa sample consisted of 28 participants with paired English and isiXhosa data (Table 1). Self-reported English proficiency was originally intended as a proficiency-band grouping variable for RQ2, but responses were heavily concentrated in the Fluent/Native categories (Table 2), leaving little variation to group on. Self-reported most comfortable language, which was more evenly distributed, was used instead. This substitution was made after inspecting the available data rather than specified in advance, which is why RQ2 is reported as exploratory throughout. Educational attainment also showed enough variation to examine, and was explored after the same inspection; it was not part of the original analysis plan and is reported separately as an exploratory appendix analysis (Appendix A.2).

3.3 SUS Instrument, Evaluated Systems, and Study Conditions

Six systems were evaluated: Google Search, Uber, EskomSePush, an automated teller machine (ATM), Takealot, and a point-of-sale (POS) machine. In Phase 1 each participant evaluated three of them using the written English SUS; in Phase 2 the intended procedure was to evaluate the other three using the written isiXhosa SUS. The language sequence itself did not vary: English was always completed before isiXhosa. Because the same participants completed both conditions, the comparison is within-participant. However, because English was always Phase 1, questionnaire language remained confounded with phase position, and within each participant the two conditions also involved different systems, so the design does not isolate a causal effect of questionnaire language. The broader SUSEVAL project used the written English condition first across all three language arms for operational consistency and to simplify coordination across phases. This decision was made for the broader project rather than for the isiXhosa component specifically. Within-subjects designs are well documented to risk practice and carryover effects when conditions are not counterbalanced [20], and the consequence here is that questionnaire language cannot be separated from phase position in any comparison reported below.

3.4 isiXhosa Translation and Adaptation

The isiXhosa SUS was developed through a cross-cultural adaptation process informed by Beaton et al. [4] and Ortiz-Gutiérrez and Cruz-Avelar [26], summarised in Figure 2 and in full in Appendix F. Two forward versions of the English SUS were produced separately: the first by the researcher with the assistance of Google Translate, the second independently by a fluent isiXhosa-speaking engineer. Before synthesis both were reviewed for linguistic, cultural, social, and production issues using categories adapted from Pan and Fond [27], and a qualified isiXhosa language specialist then compared and synthesised them into a single version.

Two independent isiXhosa–English bilingual speakers who had not seen the original SUS wording back-translated the synthesised version into English. Comparison with the original prioritised Items 3, 5, 6, and 9 for discrepancy review; Item 2 was also considered in the translator's final decision record. One back-translation reversed the polarity of Item 3, rendering "easy to use" as "hard to use." The

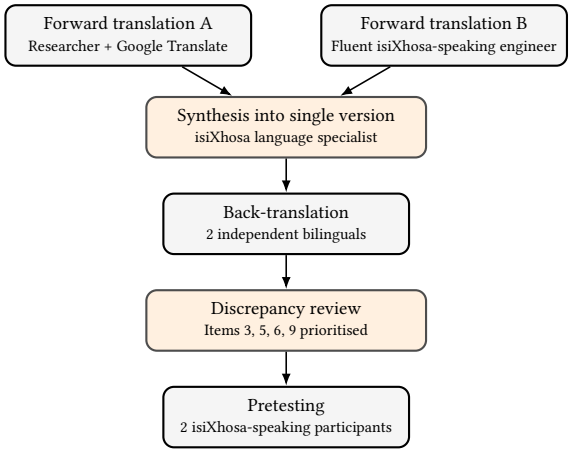


Figure 2: isiXhosa SUS adaptation process.

synthesising translator judged that the flagged discrepancies arose in back-translation and retained the synthesised wording. Because the same specialist conducted both the synthesis and the final discrepancy review, that stage should be interpreted as linguistic review rather than fully independent validation.

3.5 Pretesting

The translated SUS was pretested with two isiXhosa-speaking participants. Each evaluated one system, completed all ten isiXhosa SUS items, and provided feedback on comprehensibility and phrasing. The pretest was informed by the discrepancy review, with particular attention to Items 3, 5, 6, and 9. One pretester also completed a short follow-up concerning interpretation of the five-point response scale and use of the English wording as a reference.

3.6 Procedure, Data Collection, and Implementation Challenges

For the isiXhosa component, participants first completed the written English phase for three systems and then the written isiXhosa phase for the other three. SUS responses were submitted through Microsoft Forms without a researcher present, and demographic information was collected beforehand through the consent and demographic form. Collected data included SUS item responses, recorded system and condition, demographics, home/first language, self-reported language proficiency, educational attainment, and most comfortable language. Standard SUS scoring produced a score from 0 to 100 for each system evaluation, and the three selected system-level scores in each condition were averaged to produce participant-level English and isiXhosa scores. Optional qualitative feedback was collected after the questionnaire phases, primarily through written open-ended responses in Microsoft Forms and WhatsApp follow-up; one participant provided feedback during a WhatsApp call, for which researcher notes rather than a verbatim transcript were retained.

Protocol deviations and data-quality checks. Before analysis, original and follow-up submissions were reconciled using participant identifiers. Duplicate submissions and correction decisions were

Table 1: Data retention flow from the whole-team cleaned dataset to the primary analytic sample.

Stage	Participants	Evaluations	Note
Whole-team cleaned dataset	57	279	All study arms; not pooled into RQ1
Retained isiXhosa-arm data	29	195	Includes P025 (no Phase 1 data)
Paired sample (all retained ratings)	28	192	P025 excluded
Primary Dataset D	28	168	Non-overlapping three-system selection

Note. Analytic retention within the cleaned dataset, not the recruitment funnel.

documented, recorded system labels were checked, and unresolved protocol deviations were flagged. Follow-up responses were treated as participant-supplied corrections or supplements rather than researcher-imputed values.

The primary analysis (*Dataset D*) contained 28 paired participants, each represented by exactly three English and three isiXhosa evaluations selected to avoid same-system cross-phase overlap. One further participant with retained isiXhosa-arm data (P025) had no Phase 1 English data and could not contribute to paired analyses. Three secondary sensitivity analyses examined whether conclusions changed when: all retained evaluations for the 28 paired participants were used rather than the six-evaluation Dataset D selection; participants with any cross-phase system overlap in the retained records were excluded; and a participant with prior involvement in back-translating study materials (P014) was excluded. Table 1 reports the resulting sample sizes at the main retention stages.

Implementation challenges and procedural adaptations. Recruiting language contributors was difficult: email and online enquiries to translators and language departments produced few responses, and a telephone enquiry to the University of Fort Hare eventually produced a referral to an appropriate isiXhosa language specialist. Family and personal networks were used to identify five contributors, which allowed the adaptation to proceed but resulted in a mixture of professional and non-professional language contributors. Recruitment for the main study was also difficult, and many individuals registered interest without completing the study or responding to instructions, so the resulting sample was a purposive convenience sample that may have favoured participants who were more responsive or personally connected to the recruitment networks.

The online, unsupervised procedure also produced protocol deviations. Some participants evaluated unassigned systems, entered system names incorrectly, repeated a system, or omitted assigned systems. This disrupted the intended balance and produced a small number of cross-phase overlaps, examined through the "exclude cross-phase overlap" sensitivity analysis and, from the opposite direction, through the same-system exploratory check. Where necessary, participants were contacted for clarification or sent individualised follow-up links containing only outstanding systems; these responses came directly from participants rather than being estimated or imputed.

Separate follow-up interviews proved burdensome for some participants, so optional open-ended questions were added to the Phase 2 form, with targeted follow-up where clarification was needed. This increased qualitative participation and allowed participants to respond in the language they felt most comfortable using, but written responses offered less opportunity for probing than a semi-structured interview and varied in depth. These changes allowed data collection to continue, but reduced procedural consistency, and recruitment source, protocol deviations, follow-up completion, and qualitative response mode were documented during cleaning and considered when interpreting the findings.

3.7 Statistical Analysis Plan

SUS scores were calculated using the standard odd/even procedure [12, 21]. The three system-level scores within a language condition were averaged to give participant-level English and isiXhosa scores. Because that average combines three different systems per condition, and system allocation across conditions was uneven and not randomised (Appendix Table 7), it cannot on its own separate a language effect from a system-composition effect. Cronbach's alpha was calculated per condition from the 84 evaluations rather than the 28 participant averages; because each participant contributed three evaluations, these records are clustered within participants and the estimates are reported descriptively rather than as independent-sample validation evidence.

For RQ1 the primary analysis is an ordinary least squares model of evaluation-level SUS scores with participant and system fixed effects and participant-clustered CR1 standard errors [13], which estimates the within-person isiXhosa-minus-English difference after adjusting for additive system differences. It does not separate a language effect from a phase/order effect (Section 3.1), and with only 28 clusters cluster-robust standard errors of this kind are less reliable than with many clusters [13]. This model was adopted as primary during revision, replacing an unadjusted participant-level comparison, because that comparison cannot separate language from system composition; the concern was raised in supervisory feedback on an earlier draft. The unadjusted comparison is retained as a secondary descriptive check (paired t-test, with a Wilcoxon signed-rank test [35] and a Shapiro–Wilk check on the differences), alongside three paired re-specifications (all available evaluations for the 28 participants; excluding participants with cross-phase system overlap; excluding a participant with prior back-translation involvement) and an exploratory same-system comparison among participants who, through protocol deviations, rated an identical system in both conditions (Section 4.1.2). These are consistency checks, not independent confirmatory evidence.

RQ2 uses the same evaluation-level model extended with an interaction between questionnaire language and most comfortable language. The interaction estimates how much the within-person difference differs between participants most comfortable in English and in isiXhosa, with participant and system fixed effects. Most comfortable language does not vary within a participant, so its main effect is absorbed by the participant fixed effects; only the interaction is estimable and only it is interpreted. This specification

was also adopted during revision, after feedback that the earlier unadjusted comparison could not be interpreted. The unadjusted Welch independent-samples t-test on the difference score is retained as a secondary comparison, with a Mann–Whitney sensitivity check. The single Afrikaans-comfortable participant is excluded from both two-group comparisons, so the adjusted model uses 27 participants and 162 evaluations.

The same difference was also compared across educational attainment (High School, Undergraduate, Postgraduate) using Kruskal–Wallis, with Holm-corrected pairwise Dunn tests where the omnibus result was significant, and the Kruskal–Wallis and Welch tests were jointly Holm-corrected. This followed the same demographic inspection but was not a stated research question, so it is reported in Appendix A.2. Unlike RQ2, it was not given an evaluation-level treatment and therefore still carries the mixed-system limitation in full, which is one reason it is reported as an exploratory observation rather than a finding. Effect sizes were reported alongside significance tests [14]: Cohen's d_z for the RQ1 paired comparison, an H-based estimate for Kruskal–Wallis, Hedges' g for the comfortable-language comparison, and rank-biserial correlation for the Mann–Whitney check. Analyses were conducted in R.

3.8 Qualitative Analysis

Qualitative material was available from 14 participants. P01–P05 provided written open-ended or WhatsApp responses, except P04, whose contribution consisted of researcher notes from a WhatsApp call rather than a verbatim transcript. P06–P14 responded to optional open-ended questions included in the Microsoft Forms questionnaire. Because every qualitative question was optional, the amount and depth of material varied across participants and some responses were incomplete.

The material was analysed using reflexive thematic analysis [11]. The researcher repeatedly read the full dataset, generated initial codes focused on comprehension, perceived naturalness, ease, language preference, and explanations for response differences, and then grouped related codes into candidate themes. Themes were iteratively checked against the complete dataset and refined to retain meaningful differences and contradictory cases rather than removing them. The analysis therefore identifies patterns in the available accounts rather than estimating how prevalent each view was in the wider sample.

Participant descriptions of wording as "isiZulu-like" were treated as perceptions rather than as independent linguistic verification. P09 provided qualitative responses in isiXhosa; an English translation was reviewed with an isiXhosa-speaking collaborator who had previously assisted the study, so this translation review was not independent. Reflexively, the researcher does not speak isiXhosa, has isiZulu as a home language, and uses English predominantly in academic and formal written contexts. This position shaped the reliance on isiXhosa-speaking collaborators for language interpretation and informed attention to the distinction between home-language identity and familiarity with formal written questionnaire language.

3.9 Ethical Considerations

Ethical approval was granted by the University of Cape Town under reference SCI/02801/2026 through the Faculty of Science Research Ethics Committee. Records were linked using participant identifiers and stored in a restricted UCT Microsoft OneDrive folder, with identifying information including contact details and phone numbers kept separate from the analysis dataset. Participants completed an information sheet and consent form through Microsoft Forms before participating, and consent to participate — and, where applicable, consent to audio recording — was collected explicitly. Participation was voluntary, and participants could withdraw before anonymisation without penalty. Study communication took place primarily through WhatsApp and Microsoft Forms; participants were informed that these are third-party platforms and therefore carry some residual privacy risk. Analysis used anonymised participant IDs, and quotations reported in this paper were anonymised before inclusion. Participants were not compensated for participation.

4 Results and Discussion

4.1 Results

4.1.1 Participant Flow and Demographics. Participants completed the English version of the SUS questionnaire first and the isiXhosa version second. Table 1 reports the retention stages from the whole-team cleaned dataset to the primary sample; these counts describe analytic retention and not the complete recruitment funnel. Within the 28-participant paired sample, each participant's evaluations were restricted to three non-overlapping systems per phase, producing *Dataset D* (28 participants, 168 evaluations). The remaining 24 evaluations contribute to the "all retained ratings" sensitivity analysis and, because each completes a same-system pair with a Dataset D evaluation from the same participant, to the same-system exploratory check reported alongside it.

Table 2 summarises participant characteristics. The sample was predominantly aged 18–24 years (57.1%) and women (67.9%). Most participants reported isiXhosa as their sole home language (89.3%), with a further 10.7% reporting both isiXhosa and English as home languages. Most reported fluent (85.7%) or native (7.1%) English proficiency and native (71.4%) or fluent (28.6%) isiXhosa proficiency. Half (50.0%) identified English as their most comfortable language, 46.4% isiXhosa and one participant (3.6%) Afrikaans.

4.1.2 RQ1: English and isiXhosa SUS Scores. RQ1 asked how SUS scores differed between the written English and written isiXhosa questionnaire conditions. This is addressed first with the evaluation-level model that is the primary analysis (Section 3.7), then with the unadjusted participant-level comparison reported for consistency with prior SUS studies.

Descriptive statistics and reliability. Table 3 reports participant-level descriptive statistics ($n = 28$ participant averages per version). Internal consistency was computed across the 84 evaluations contributed to each version by the same 28 participants, not across the 28 participant averages. These refer to different units of analysis and are therefore reported separately. A graphical summary is in Appendix Figure 5.

Table 2: Participant characteristics (N = 28).

Variable	Category	n	%
Age	18–24	16	57.1
	25–34	2	7.1
	35–44	4	14.3
	45–54	5	17.9
	55+	1	3.6
Gender	Woman	19	67.9
	Man	9	32.1
Education	High School	11	39.3
	Undergraduate degree/diploma	6	21.4
	Postgraduate degree	11	39.3
Home language	isiXhosa	25	89.3
	isiXhosa and English	3	10.7
English proficiency	Fluent	24	85.7
	Native	2	7.1
	Conversational	2	7.1
isiXhosa proficiency	Native	20	71.4
	Fluent	8	28.6
Most comfortable language	English	14	50.0
	isiXhosa	13	46.4
	Afrikaans	1	3.6

Table 3: SUS scores and internal consistency by questionnaire language.

Version	Mean SUS[a]	SD[a]	Cronbach's α[b]
English	74.85	9.81	.843
isiXhosa	75.48	11.56	.890

[a] 28 participant-level averages per language. [b] 84 evaluations per language.

Both versions showed good internal consistency ($\alpha > .80$). This reflects the SUS questionnaire's internal structure in this sample and evaluation set; it is not evidence of measurement equivalence between the two language versions.

Primary comparison: evaluation-level model. The primary test of RQ1 was the OLS model of evaluation-level SUS scores with participant and system fixed effects and participant-clustered CR1 standard errors. It estimated the within-person isiXhosa-minus-English difference as 0.97 SUS points (95% CI [−4.38, 6.32], $p = .713$). The interval includes zero, so the model did not detect a difference between conditions after adjusting for system. Because English was always Phase 1, this estimate still reflects questionnaire language and phase position jointly rather than isolating either one.

Secondary unadjusted comparison: participant-level condition means. For comparison with prior SUS reporting conventions, Table 3 also reports participant-level means: English $M = 74.85$ ($SD = 9.81$), isiXhosa $M = 75.48$ ($SD = 11.56$), unadjusted mean within-person difference = 0.62 points (95% CI [−4.63, 5.88]), paired $t(27) = 0.244$, $p = .809$, $d_z = 0.046$; Wilcoxon $V = 216.5$, $p = .516$; Shapiro–Wilk $W = 0.942$, $p = .123$. This is directionally consistent with the system-adjusted estimate but combines different systems per condition, so it is not independent confirmatory evidence. Individual differences varied widely (SD = 13.55); per-participant means are in Appendix Table 11. Figure 3 shows the participant-level change; the distribution of differences is in Appendix Figure 6.

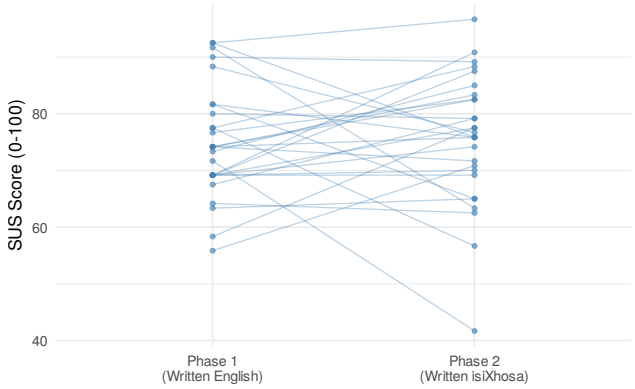


Figure 3: Participant-level SUS scores across questionnaire conditions.

Note. Each line represents one participant. Each endpoint is the mean of three systems; the systems differ between conditions.

Same-system exploratory check. Protocol deviations meant 10 of the 28 paired participants rated the same system in both conditions, contributing 24 same-system pairs (48 evaluations) across all six systems, each pairing one Dataset D evaluation with one retained evaluation from outside it (Appendix Table 12). This subset was unplanned and too small to be conclusive alone, but is the closest available same-system comparison. Averaging each participant's overlapping-system evaluations, the mean isiXhosa-minus-English difference was 2.40 points ($SD = 10.49$, 95% CI [−5.11, 9.90], $t(9) = 0.72$, $p = .488$; Wilcoxon $V = 35$, $p = .474$), consistent with both estimates above.

The participant with the largest change, P053 (Figure 3, Appendix Table 11), showed a −30.00 point assigned-system difference. Restricting the comparison to the three systems they rated in both conditions reduced this to −8.33 points. This suggests that system composition may have contributed to the change, but three pairs are too few to establish that. P057, the participant with the second-largest change (−28.33), had no same-system overlap, so this check cannot explain their change.

System-level description. The realised number of evaluations per system differed between conditions within Dataset D, so per-system means should not be read as language-effect estimates; full counts are in Appendix Table 7.

Robustness checks. All six estimates, the primary system-adjusted model, the unadjusted comparison, three alternative paired specifications and the same-system check, are small and non-significant with confidence intervals including zero; full results are in Appendix Table 6.

4.1.3 RQ2: Most Comfortable Language. RQ2 examines whether the within-person English–isiXhosa SUS difference varies with participants' most comfortable language. This characteristic was substituted for an originally planned proficiency-based grouping

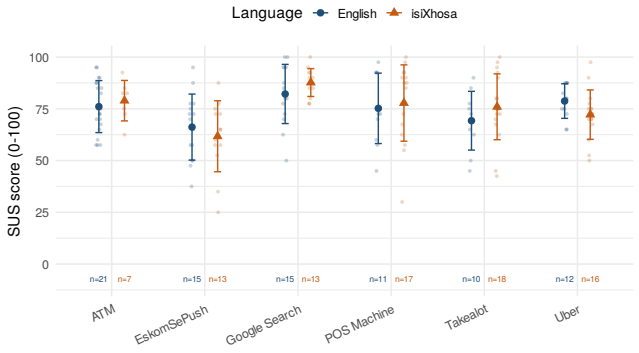


Figure 4: SUS scores by system and questionnaire language.

Note. Small points are individual evaluations; large markers are means; error bars show ±1 SD; *n* gives evaluation counts. Comparisons are descriptive because evaluation counts and participant composition within systems differ between conditions.

after inspection of the available data, as described in the Methods; the analysis is exploratory and the characteristic was not manipulated, so no causal claims are made. RQ2 is addressed by the system-adjusted interaction model (Section 3.7), reported first below; the unadjusted between-group comparison of participant-level difference scores follows it as a secondary check and carries the mixed-system limitation of the unadjusted RQ1 comparison. Educational attainment was also examined after the same demographic inspection but was not part of the original analysis plan; it is reported as an exploratory appendix analysis (Appendix A.2) rather than as an answer to RQ2.

Most comfortable language. Table 4 reports descriptive statistics by most comfortable language. The single Afrikaans-comfortable participant is descriptive only and was excluded from the inferential comparison below.

Table 4: SUS score differences by most comfortable language.

Group	n	Mean diff.	SD	Median	IQR
English	14	2.14	15.45	8.75	20.83
isiXhosa	13	−1.09	12.18	0.83	7.50
Afrikaans (descriptive only)	1	1.67	—	1.67	—

Note. Difference = isiXhosa minus English. These are unadjusted participant-level descriptive differences, not estimates from the system-adjusted interaction model.

The system-adjusted interaction model (27 participants, 162 evaluations) estimated the language-by-comfort interaction at 2.43 SUS points (95% CI $[-8.21, 13.07]$, $t(26) = 0.47$, $p = .643$): with participant and system fixed effects, the within-person isiXhosa-minus-English difference was 2.43 points higher for English-comfortable than for isiXhosa-comfortable participants, with an interval spanning roughly 21 points and including zero. The model therefore did not detect a subgroup difference. Because English always preceded isiXhosa, this interaction reflects language together with phase position as the RQ1 estimate does; adjusting for system composition does not recreate a controlled comparison or remove the fixed-order confound.

The unadjusted comparison agrees. A Welch test on the English-comfortable ($n = 14$) and isiXhosa-comfortable ($n = 13$) groups' mean within-person differences gave 3.23 points (95% CI [−7.77, 14.23]), $t(24.39) = 0.606$, $p = .550$ (Holm-adjusted $p = .550$), Hedges' $g = 0.22$, with a Mann–Whitney sensitivity check consistent with it ($U = 112.5$, $p = .308$, rank-biserial $r = 0.236$). The adjusted estimate is the main result because it accounts for additive system differences at the evaluation level. These null results should not be read as evidence of no association. With only 14 and 13 participants, the interval remains compatible with a substantial association in either direction.

4.2 Qualitative Findings

The qualitative analysis examined how participants experienced the English and isiXhosa versions of the SUS, with particular attention to what shaped comprehension, ease of answering, and language preference. Three themes were developed from the participant accounts; Table 9 in Appendix A.1 summarises them. Because English always preceded isiXhosa and participants evaluated different systems across the two phases, comments about differences in responses are treated as participants' explanations of their experiences rather than as evidence of an isolated language effect.

Familiarity and exposure often made English easier. Participants who found English easier frequently explained this in terms of previous exposure rather than describing English itself as inherently clearer. P08 connected English ease to being "more familiar with using english when interacting with technology," while P02 referred both to familiarity with English technology terms and to greater experience completing surveys in English. Similarly, P10 attributed the difference to greater exposure to English questionnaires, and P03 explained that they did not usually read in isiXhosa.

Home-language status alone therefore did not determine which written questionnaire felt easiest. P13 provided a contrasting case, finding isiXhosa easier because it was their first language, while P14 reported both versions equally easy because they were fluent in both. P01, P06 and P07 also preferred English, but for reasons relating to the wording and register of the translation, considered under the second theme.

Comprehension and naturalness were not the same thing. A second pattern was that understanding a translated question and finding its wording natural were not necessarily the same judgement. P01 described some isiXhosa wording as "very deep" and unlike everyday conversation, while P12 similarly observed that people did not normally use that kind of language in everyday speech and instead often mixed isiXhosa and English. P07 described the isiXhosa questionnaire as visibly feeling like a translation, while still reporting that it made sense.

This concern was not unique to isiXhosa. P02 described the English expressions "unnecessarily complex" and "cumbersome" as unusually formal, and P14 reported encountering the word "cumbersome" for the first time and searching for its meaning before answering. This suggests that some difficulties concerned the register of the SUS itself rather than translation alone.

Participants differed in whether translation-like wording was problematic. P08 noticed wording that felt directly translated from

English but described this as normal in isiXhosa and reported the questions remained understandable. In contrast, P05 said unfamiliar translated words affected whether the two versions appeared to ask the same thing and described becoming "lost in translation," and P06 answered "No" when asked whether the two versions appeared to ask the same thing. Comprehension was therefore not always preserved despite unnatural wording. P06 and P09 additionally perceived some wording as resembling isiZulu; these are treated as participants' perceptions rather than independent linguistic verification.

Ease and language preference did not always align. For some participants, the easiest version was not the one they preferred. P02 gave the clearest example, preferring English for speed but isiXhosa for "comfort and deep understanding." P10 also found English easier because of greater exposure to English questionnaires, but preferred isiXhosa because they felt that insufficient research was conducted in their home language. P14 similarly preferred isiXhosa and linked this to having relatively few opportunities to use the language.

P09 reported English as easier while preferring isiXhosa; no explanation was recorded and none is inferred here. Conversely, P13 showed ease and preference could align, isiXhosa being both easier and preferred as their first language. Most other participants who preferred English also found English easier, so the separation between ease and preference was meaningful but not universal.

Additional instrument-level observations. Some feedback concerned the SUS items themselves rather than either language version: P02 questioned whether the item on support from a technical person referred to initial or repeated use, and felt the complexity and inconsistency items appeared similar. Uncertainty may therefore arise from the concepts and wording of the SUS itself rather than from translation alone.

Participants who answered the two phases differently frequently attributed this to contextual factors rather than to language. P01 and P07 referred to evaluating different systems, and P13 to their experience with Uber. P02 gave a different explanation: answering in English they imagined people their own age, whereas the isiXhosa wording prompted them to think about older relatives, affecting their rating. P01 also said that completing English first helped them understand the subsequent isiXhosa questions and that reversing the order might have made some items less clear, consistent with the general finding that responses can be shaped by what preceded them [30], though this design cannot establish that a carryover effect occurred. P09 and P07 suggested the time between phases could also have affected responses. These accounts reinforce the need to avoid attributing score differences solely to questionnaire language and support treating the English-first procedure as a design limitation.

4.3 Discussion

4.3.1 English and isiXhosa SUS Performance. Across the system-adjusted primary model, the unadjusted paired comparison, three further paired re-analyses, and the same-system exploratory check, the estimated isiXhosa-minus-English difference remained small (from −0.15 to +2.40 SUS points), and each confidence interval included zero. Because questionnaire language and phase position were fully confounded, none of these analyses can identify a language effect in isolation; together, they show that the near-zero pattern was not materially changed by how system composition was handled. The primary hypothesis was therefore not supported in this sample. Participants' accounts also show why the differences are hard to attribute to language alone: P01 and P07 attributed within-person differences to evaluating different systems rather than to language, and P01 reported that completing the English questionnaire first helped them understand the subsequent isiXhosa items (Section 4.2). System composition and phase order are therefore both plausible contributors to the observed heterogeneity alongside language, and this design cannot separate their contributions. The qualitative accounts also help explain why the average difference was small. Participants who found English easier, such as P08, P02 and P10, attributed this to familiarity with written English, technology terminology and prior experience of English questionnaires rather than to English being clearer in itself, while P14 reported both versions equally easy and P13 found isiXhosa easier as their first language. These are explanations of comprehension and ease, not of preference, and they should not be read as evidence of a learning effect or of measurement equivalence; P05 and P06 reported that unfamiliar translated wording did interfere with whether the two versions appeared to ask the same thing.

The internal-consistency estimates for both conditions were above .80 and are broadly in the range reported for several translated SUS versions in prior work [16, 21]. However, the present alpha estimates pool repeated evaluations from the same participants and do not establish linguistic or psychometric equivalence. More generally, a near-zero mean difference should not be interpreted as evidence that participants experienced the two questionnaires identically: the confidence interval around the primary estimate remains compatible with modest differences in either direction, and the individual difference scores were widely dispersed (SD ≈ 13.5 points).

For RQ2, the system-adjusted interaction model (27 participants, 162 evaluations) estimated the language-by-comfort interaction at 2.43 SUS points (95% CI $[-8.21, 13.07]$, $t(26) = 0.47$, $p = .643$). The isiXhosa-minus-English difference was therefore estimated to be 2.43 points higher among English-comfortable participants than among isiXhosa-comfortable participants, but the confidence interval was wide and included zero. The model therefore did not detect a subgroup difference. As with RQ1, questionnaire language remained confounded with phase position, and adjusting for system does not remove that limitation. An exploratory comparison by educational attainment is reported in Appendix A.2.

Overall, the quantitative results show no statistically detectable average difference between the two conditions, while leaving open substantial individual heterogeneity that this design cannot attribute specifically to language, system, or order. Separating those contributions would require a larger counterbalanced study. These findings are most informative about highly proficient bilingual isiXhosa–English speakers, the profile of most participants here (Table 2), and should not be generalised to isiXhosa speakers with limited English proficiency.

4.3.2 Language Familiarity, Register, and Preference. The quantitative and qualitative findings together suggest that questionnaire language cannot be reduced to a simple first-language-versus-English distinction. Although participant-level means were similar, English was often described as easier because it was familiar from questionnaires, technology use and education, while isiXhosa could still be preferred for comfort, home-language identification or representation.

A participant's home language, easiest written language and preferred questionnaire language were therefore not necessarily the same, echoing cross-cultural adaptation guidance that distinguishes literal wording from semantic, experiential and conceptual equivalence [4, 26]. Participants' descriptions of isiXhosa wording as formal or translation-like, alongside reports of habitually mixing isiXhosa and English in everyday speech, are consistent with a broader distinction between named, standardised written varieties and the fluid everyday practices multilingual speakers actually use [17]; a South African case study of Grade 6 isiXhosa home-language writing documents a similar mismatch between the everyday and urban varieties students bring to the classroom and the standardised variety expected in formal assessment [24]. Translation register and experience reading formal isiXhosa therefore matter alongside first-language identity.

Two possibilities are worth testing in future work. A bilingual presentation could let respondents consult both wordings when one register feels unfamiliar, and an orally administered isiXhosa SUS could test whether written register contributes to difficulty. Neither was evaluated here, and a bilingual format would be unsuitable for a study attempting to isolate questionnaire-language effects, so both are directions for future testing rather than demonstrated improvements.

4.3.3 Translation as a Starting Point, Not an Endpoint. P04 raised a broader issue by connecting questionnaire accessibility with difficulties using technologies such as Uber and ATMs, literacy in both English and isiXhosa, and risks when users require help with unfamiliar systems. A single account cannot establish how common these problems are, but it illustrates the distinction between making usability *measurement* linguistically accessible and making the underlying *technology* linguistically accessible. The adaptation process undertaken here addresses the wording of the SUS; it cannot by itself make the evaluated systems available or natural to use through isiXhosa. Translation is therefore a starting point for inclusive usability evaluation rather than evidence that the wider interaction context has been localised.

5 Limitations and Future Work

Several limitations constrain the interpretation of these findings. First, English always preceded isiXhosa, so questionnaire language was fully confounded with phase position, and within each participant the two phases involved different systems whose realised distribution remained uneven (Section 3.1). The observed within-person difference therefore cannot be interpreted as an isolated language effect. The unsupervised online procedure also produced incomplete submissions, incorrect system labels, duplicate ratings, and some cross-phase overlap; follow-up responses and sensitivity analyses reduced their impact but could not restore a controlled or balanced design.

Second, the primary paired sample was small ($N = 28$) and recruited purposively. The RQ2 grouping variable was substituted after inspecting the demographic distributions, so the RQ2 finding is exploratory. The system-adjusted interaction accounts for additive system differences under the model's assumptions but does not remove the design limitations or address the fixed language order; with only 14 and 13 participants in the two groups, it is also imprecise. The educational-attainment groups in Appendix A.2 were small, unequal and overlapped unevenly with most comfortable language, so that comparison is even less conclusive.

Third, the adaptation process combined professional and non-professional contributors: the first forward version was machine-assisted, the second was produced by a fluent isiXhosa speaker who was not a professional translator, pretesting involved only two participants, and the same specialist synthesised the forward versions and reviewed the back-translation discrepancies. P09's isiXhosa response translation was also reviewed by a collaborator who had already assisted the study. These procedures provided useful linguistic checks but do not constitute independent validation.

Finally, qualitative material was collected through mixed modes and varied substantially in depth: most responses were optional written answers, while one participant was represented by researcher notes from a WhatsApp call rather than a transcript. The study therefore does not establish measurement invariance, full psychometric validity, or equivalence between the English and isiXhosa SUS.

Future work should use a counterbalanced design [20] in which questionnaire language order is varied and the same-system comparison is controlled more directly. Researcher-supervised sessions or a single structured survey could reduce protocol deviations and attrition. Larger, more systematically recruited samples would support more stable subgroup analyses and fuller psychometric evaluation. Future studies could also compare written and orally administered isiXhosa SUS versions and use trained bilingual interviewers so that qualitative interviews can be conducted in participants' preferred language and translated through a documented independent process.

6 Conclusion

Among 28 paired participants, neither the system-adjusted model nor the unadjusted participant-level comparison found a statistically detectable average difference between the written English and written isiXhosa conditions. Internal consistency was good in both. Because English always came first and participants evaluated different systems across phases, the study does not isolate a questionnaire-language effect. The exploratory interaction model also did not detect a difference by most comfortable language, although the confidence interval was wide enough that a substantial association cannot be ruled out. These findings are most informative about highly proficient bilingual isiXhosa–English speakers and should not be generalised more broadly. Qualitative accounts showed that English familiarity, written register, and language preference shaped participants' experiences in ways the aggregate scores did not capture.

The value of an isiXhosa SUS may extend beyond whether it produces a different numerical score from the English version. Developing the adaptation highlighted a broader problem: making a usability questionnaire available in isiXhosa does not necessarily make either the questionnaire or the technology being evaluated fully accessible to isiXhosa speakers. Participants' experiences point to the importance of written-language familiarity, everyday language practices, digital literacy, and the language of the underlying system.

Translation is an important step towards more inclusive usability evaluation, but it is not the endpoint. Future work should ask not only whether the SUS can be translated into isiXhosa, but also whether the digital systems being evaluated can themselves be used comfortably through isiXhosa.

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A Additional Quantitative Results

Table 5: Selected reported psychometric findings for SUS translations.

Language and study	n	α	Reported Validity
Arabic [1]	90	.82	NR
Slovene [5]	182	.81	.52
Polish [7]	NR	.81	.82
Italian [8–10]	20–196	.81–.84	.45–.95
Persian [15]	202	.79	NR
Portuguese [23]	32	NR	.70

Note. NR = not reported in the cited study(s). Validity coefficients are reported as presented in the source studies and are not directly comparable because different validation approaches and reference measures were used.

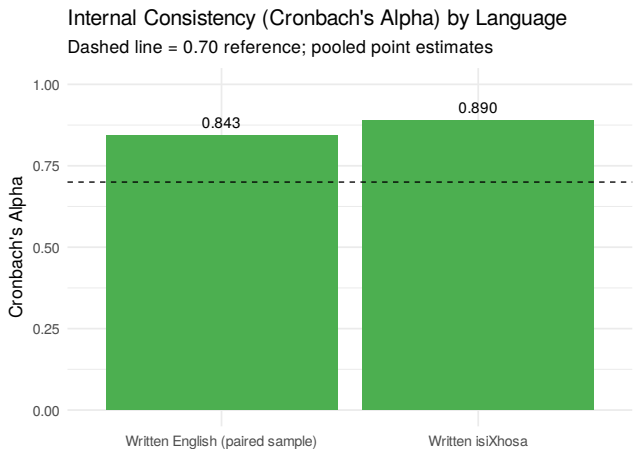


Figure 5: Internal consistency by questionnaire language.

Note. Each estimate pools 84 evaluations from 28 participants.

A.1 Qualitative Theme Summary

Table 9 summarises the three themes developed in Section 4.2; the themes and their contrary cases are described in full there.

A.2 Educational Attainment (Exploratory)

Educational attainment was examined after inspecting the collected demographic distributions (Section 3.2), rather than planned in advance, and is not one of the paper's stated research questions; it is reported here as an exploratory observation, separate from RQ2. Table 10 shows the within-person SUS difference by group. A Kruskal–Wallis test produced $H(2) = 7.69$, $p = .021$ ($\eta^2_H = 0.228$, a standardised estimate rather than a directly interpretable variance proportion); the Holm correction described in Section 3.7 was applied across two tests, this Kruskal–Wallis test and the unadjusted Welch comparison of most comfortable language, giving $p = .043$ for education and $p = .550$ for the Welch test. That correction family predates the system-adjusted interaction model reported in Section 4.1.3 and does not include it.

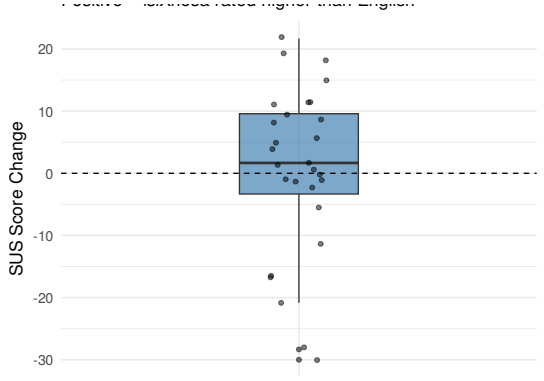


Figure 6: Distribution of within-participant SUS score differences.

Note. Positive values indicate higher scores in the isiXhosa condition. Differences compare participant means across different systems in the two conditions; see Section 4.1.2 for the system-adjusted primary estimate.

Pairwise Dunn tests (Table 8) identified only the Undergraduate–Postgraduate contrast as significant after Holm correction ($z = -2.72$, $p_{\text{Holm}} = .020$; both High School contrasts $p_{\text{Holm}} = .205$): Undergraduate participants scored higher in isiXhosa, Postgraduate participants higher in English, High School in between. This is best treated as hypothesis-generating, not confirmatory: it was not a stated research question, was selected post-hoc, includes only six Undergraduate participants, uses the same mixed-system averages as RQ2 without an evaluation-level treatment, and overlaps unevenly with most comfortable language (8 of 11 High School participants were English-comfortable; 7 of 11 Postgraduate participants isiXhosa-comfortable), so the two comparisons cannot isolate independent associations.

A.3 Participant-Level Condition Means and Differences

Table 11 lists each participant's condition means and their signed difference, ordered by absolute difference. It supports the individual-variation and outlier discussion in Section 4.1.2.

A.4 Same-System Pairs

Table 12 lists the 24 same-system pairs underlying the exploratory check in Section 4.1.2, grouped by participant.

A.5 Distribution of Participant-Level Differences

Figure 6 shows the distribution of the within-participant differences summarised in Section 4.1.2; the participant-level comparison itself is in Figure 3. Both are descriptive and carry the mixed-system caveat noted in their captions.

Table 6: RQ1 primary and secondary comparisons.

Analysis	n	Eval.	Estimate	95% CI	p
System-adjusted model (primary)[d]	28	168	0.97	[−4.38, 6.32]	.713
Unadjusted paired comparison (Dataset D)	28	168	0.62	[−4.63, 5.88]	.809
All retained ratings[a]	28	192	0.85	[−3.74, 5.44]	.707
Excl. cross-phase overlap[b]	18	108	1.44	[−4.75, 7.62]	.631
Excl. P014[c]	27	162	−0.15	[−5.36, 5.05]	.952
Same-system exploratory check[e]	10	48	2.40	[−5.11, 9.90]	.488

Note. Estimates are isiXhosa minus English, in SUS points; n counts participants.
[a] All retained ratings from the same 28 participants.
[b] Excludes participants with any cross-phase system overlap in retained records.
[c] Excludes a participant involved in back-translation.
[d] Participant and system fixed effects with participant-clustered CR1 standard errors.
[e] 24 same-system pairs (48 evaluations) among the 28 paired participants, arising from protocol deviations rather than the intended design; participant-level average of overlapping-system differences.

Table 7: System-level SUS scores by questionnaire language.

Version	System	n[a]	Mean SUS	SD
English	ATM	21	76.07	12.56
English	EskomSePush	15	66.17	15.95
English	Google Search	15	82.17	14.29
English	POS Machine	11	75.23	17.01
English	Takealot	10	69.25	14.19
English	Uber	12	78.75	8.36
isiXhosa	ATM	7	78.93	9.77
isiXhosa	EskomSePush	13	61.73	17.15
isiXhosa	Google Search	13	87.69	6.73
isiXhosa	POS Machine	17	77.79	18.43
isiXhosa	Takealot	18	75.97	15.91
isiXhosa	Uber	16	72.19	11.93

[a] Number of evaluations. Comparisons are descriptive; realised system counts and participant composition within systems differ across questionnaire conditions.

Table 8: Pairwise Dunn tests for educational attainment.

Comparison	z	p[a]
High School vs. Undergraduate	1.347	.205
High School vs. Postgraduate	−1.634	.205
Undergraduate vs. Postgraduate	−2.720	.020

[a] Holm-adjusted across three comparisons.

B Study Materials

B.1 Systems and Tasks

Participants evaluated six everyday systems. The interaction requested for each system is summarised in Table 13.

C Participant Instructions

Participants received the following instructions at the beginning of the study. The instructions explained the two-phase structure of the study, the tasks to be completed, and the follow-up procedure.

C.1 Initial Group Message

The following message was pinned in the community/group channel and was also included in the group information:

> **Welcome! Thanks for joining the SUSEVAL usability study.**
> **Step 1:**
> Complete the short consent + demographic form:
> English: English consent and demographic form
> isiXhosa: isiXhosa consent and demographic form
> Afrikaans: Afrikaans consent and demographic form
> **Step 2:**
> A group admin will DM you directly with your personalised instructions. You will not need to do anything else in this group chat.
> **Privacy:**
> Your number stays private from other participants.
> All study communication happens one-on-one.
> Any questions? Just message a group admin directly.

Equivalent versions of this message were provided in Afrikaans and isiXhosa for participants who preferred those languages.

C.2 Daily Group Reminder

A reminder message was sent daily in the community/group:

> Hello everyone, thank you for joining. Please complete the consent and demographic form in your preferred language if you haven't already. An admin will contact you personally afterwards (within 24 hours) with further instructions. If you've already done so, please check your direct messages.
> English: English consent and demographic form
> isiXhosa: isiXhosa consent and demographic form
> Afrikaans: Afrikaans consent and demographic form

C.3 Individual Participant Instructions

After a participant completed the consent and demographic form, a group administrator contacted the participant directly. The following message was used to explain the study procedure:

> **Thanks for joining the SUSEVAL study! Here's how it will work.**
> You'll be evaluating **6 everyday apps/tasks** in two stages. There are no right or wrong answers. We're interested in your honest experience.

Table 9: Summary of themes identified in participant accounts of the English and isiXhosa SUS.

Theme	Core idea	Supporting participants	Contrasting case(s)
Familiarity and exposure often made English easier	Participants often explained greater ease with English through familiarity with written English, technology terminology, questionnaires, or everyday reading rather than through English being inherently easier.	P02, P03, P08, P10	P13 found isiXhosa easier because it was their first language; P14 found both equally easy.
Comprehension and naturalness were not the same thing	Several participants reported understanding the isiXhosa questions while still describing their wording as formal, lengthy, unfamiliar, or noticeably translated. However, for some participants unfamiliar wording did interfere with perceived equivalence or answering.	P01, P02, P07, P08, P12	P05 reported being "lost in translation"; P06 did not perceive the two versions as equivalent.
Ease and language preference did not always align	For some participants, the version that was quickest or easiest to answer was not necessarily the version they would prefer to use. Preference was also shaped by comfort, home-language use, and representation.	P02, P10, P14	P09 preferred isiXhosa despite finding English easier, but gave no reason; P13's ease and preference both favoured isiXhosa.

Table 10: SUS score differences by educational attainment (exploratory).

Group	n	Mean diff.	SD	Median	IQR
High School	11	2.50	10.92	5.00	9.58
Undergraduate degree/diploma	6	10.97	9.71	11.25	8.33
Postgraduate degree	11	−6.89	14.10	−0.83	17.50

Note. Difference = isiXhosa minus English.

Table 11: Participant-level condition means and differences, ordered by absolute difference.

Participant	English	isiXhosa	Difference
P053	71.67	41.67	−30.00
P057	91.67	63.33	−28.33
P014	69.17	90.83	+21.67
P047	77.50	56.67	−20.83
P054	58.33	77.50	+19.17
P019	69.17	87.50	+18.33
P021	92.50	75.83	−16.67
P056	81.67	65.00	−16.67
P052	55.83	70.83	+15.00
P029	67.50	79.17	+11.67
P012	73.33	85.00	+11.67
P016	88.33	76.67	−11.67
P051	77.50	88.33	+10.83
P004	74.17	83.33	+9.17
P038	74.17	82.50	+8.33
P042	69.17	77.50	+8.33
P001	81.67	75.83	−5.83
P013	76.67	82.50	+5.83
P027	69.17	74.17	+5.00
P015	92.50	96.67	+4.17
P020	74.17	71.67	−2.50
P039	64.17	62.50	−1.67
P024	63.33	65.00	+1.67
P049	74.17	75.83	+1.67
P034	69.17	70.00	+0.83
P046	90.00	89.17	−0.83
P050	80.00	79.17	−0.83
P036	69.17	69.17	+0.00

Note. Dataset D ($n = 28$). Each condition mean averages three different systems, so the two means are not matched on system. Difference = isiXhosa minus English; positive values indicate higher scores in the isiXhosa condition. Values are shown to two decimal places; ordering and all calculations use the unrounded values. The ordering describes observed within-person change and does not isolate a questionnaire-language effect.

Phase 1: Written English (3 tasks)
For each of these, you'll be given three short tasks, then asked to fill in a 10-question form about the experience for each task. It should take about 5 minutes each. If a task involves a physical thing (like an ATM or point-of-sale (POS) machine), just try it next time you naturally would, then fill in the form afterwards. No need to do it on the spot.
Phase 2: Written isiXhosa (3 tasks)
Once I've received your Phase 1 responses, I'll follow up shortly after with Phase 2. It is the same idea, but this time the questionnaire will be in isiXhosa.
Follow-up questions
Once you've finished both phases, I may reach out to ask if you'd like to share a bit more about your experience, such as through a short voice note or call. This would take approximately 5–10 minutes and would focus on what felt easy or confusing.
Everything you share stays confidential and is only used for this research. You can stop at any point with no penalty. Let me know if anything's unclear before we start.

C.4 System Allocation Across Study Phases

The six evaluated systems were distributed across the two study phases. Participants were intended to evaluate three systems during Phase 1 using the written English SUS and the remaining three systems during Phase 2 using the written isiXhosa SUS. The researcher

Table 12: Same-system pairs arising from protocol deviations, grouped by participant.

Participant	System	English	isiXhosa	Difference	D
P001	ATM	87.50	72.50	−15.00	X
	EskomSePush	82.50	77.50	−5.00	X
P004	ATM	75.00	97.50	+22.50	E
	EskomSePush	77.50	82.50	+5.00	E
	Google Search	70.00	80.00	+10.00	E
P013	ATM	57.50	70.00	+12.50	E
	Takealot	90.00	92.50	+2.50	E
	Uber	82.50	100.00	+17.50	E
P014	ATM	87.50	92.50	+5.00	X
	Google Search	90.00	87.50	−2.50	X
	Takealot	72.50	75.00	+2.50	E
	Uber	87.50	80.00	−7.50	E
P015	ATM	87.50	97.50	+10.00	E
P016	ATM	85.00	77.50	−7.50	E
	EskomSePush	87.50	75.00	−12.50	E
P038	EskomSePush	50.00	50.00	+0.00	E
	Google Search	85.00	87.50	+2.50	E
P047	ATM	87.50	75.00	−12.50	E
	EskomSePush	70.00	77.50	+7.50	E
	POS Machine	75.00	75.00	+0.00	E
P051	Uber	75.00	95.00	+20.00	E
P053	ATM	67.50	55.00	−12.50	E
	Google Search	82.50	65.00	−17.50	E
	Takealot	65.00	70.00	+5.00	E

Note. 24 pairs across 10 of the 28 paired participants, involving 48 evaluations. Every pair contains exactly one evaluation retained in Dataset D and one from the additional retained evaluations outside it; the **D** column records which side is the Dataset D evaluation (E = the English evaluation, X = the isiXhosa evaluation). Difference = isiXhosa minus English. The overall same-system estimate reported in Section 4.1.2 averages the pair differences within each participant first and then across the 10 participants, so it is not the mean of these 24 rows. These overlaps arose from protocol deviations rather than from a planned same-system design and are not a controlled comparison.

Table 13: Systems evaluated and task instructions.

System	Task instruction
Uber	Get to the fare estimate screen for a trip of your choice; booking was not required.
Takealot	Search for a product and add it to the cart or wishlist.
EskomSePush	Add an area and check its current load-shedding schedule.
ATM	When next withdrawing cash, pay attention to the interaction.
Google Search	Search for a topic of the participant's choice.
POS machine	When next paying by card, pay attention to the interaction.

manually varied which systems were assigned to each phase across participants, with the practical aim of keeping system representation reasonably balanced across the sample. Assignments were not generated from a formal randomisation or allocation schedule.

Participants were instructed to rate only their assigned systems and not to repeat a system in Phase 2 if it had already been evaluated in Phase 1. Recruitment and completion patterns, protocol deviations, and incomplete responses meant that the observed distribution was not balanced in practice. System-level counts for the primary analytic dataset are reported in Table 7.

C.5 Phase 1 Direct Message

The final Phase 1 message used to clarify the assigned systems was:

> The 3 systems you will be evaluating in English are (Please rate ONLY these systems):
> (1) Uber
> (2) Takealot
> (3) POS Machine (the card reader or Yoco you use at stores to pay with)
> Please let me know once you have completed that.

This clarification was introduced to ensure that participants rated only their assigned systems and did not inadvertently complete questionnaires for the other systems.

C.6 Phase 2 Direct Message

After completion of Phase 1, participants received the Phase 2 questionnaire and the following instructions:

> **Thanks for completing Phase 1! Here's Phase 2:**
> Same idea as before, but this time for these 3 tasks, in isiXhosa:
> **NB:** It is very important for Part 2 that you do not rate systems you already rated in Part 1.
> (1) Google Search engine
> (2) An ATM
> (3) EskomSePush
> Take your time with each. If any word or question feels unclear, just ask me directly.
> The Phase 2 questionnaire was provided through the Phase 2 questionnaire.

D Questionnaire Wording

Both questionnaire conditions used the ten SUS items with a five-point agreement response scale. Phase 1 used the standard English SUS wording described by Brooke [12]. Phase 2 used the final synthesised isiXhosa wording developed in this study.

D.1 English SUS Wording

The English condition used the standard ten-item SUS wording described by Brooke [12]. The exact form administered to participants is reproduced in Figure 7.

D.2 Final isiXhosa SUS

(1) Ndicinga ukuba ndingathanda ukuyisebenzisa rhoqo le nkqubo.

(2) Ndiyifumanise inzima ngokungeyomfuneko le nkqubo.

(3) Ndicinga ukuba bekulula ukuyisebenzisa le nkqubo.
(4) Ndicinga ukuba ndiza kudinga uncedo lomntu onobugcisa ukuze ndikwazi ukuyisebenzisa le nkqubo.
(5) Ndifumanise ukuba imisebenzi eyohlukeneyo kule nkqubo idityaniswe kakuhle.
(6) Bendicinga ukuba kukho izinto ezininzi ezingahambelaniyo kule nkqubo.
(7) Ndicinga ukuba uninzi lwabantu lungafunda ukuyisebenzisa ngokukhawuleza le nkqubo.
(8) Ndikufumanise kunzima ukuyisebenzisa le nkqubo.
(9) Ndizive ndinokuzithemba okukhulu ukuyisebenzisa le nkqubo.
(10) Kuye kwafuneka ukuba ndifunde izinto ezininzi phambi kokuba ndiqalise ukuyisebenzisa le nkqubo.

D.3 Original Administered Forms

Figures 7–9 reproduce the questionnaire forms used during data collection.

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1. I think that I would like to use this system frequently	○	○	○	○	○
2. I found the system unnecessarily complex	○	○	○	○	○
3. I thought the system was easy to use	○	○	○	○	○
4. I think that I would need the support of a technical person to be able to use this system	○	○	○	○	○
5. I found the various functions in this system were well integrated	○	○	○	○	○
6. I thought there was too much inconsistency in this system	○	○	○	○	○
7. I would imagine that most people would learn to use this system very quickly	○	○	○	○	○
8. I found the system very cumbersome to use	○	○	○	○	○
9. I felt very confident using the system	○	○	○	○	○
10. I needed to learn a lot of things before I could get going with this system	○	○	○	○	○

Figure 7: Written English SUS form used in Phase 1.

Kwisitshixo ngasinye kwezi zilandelayo, phawula ibhokisi enye echaza kakuhle impendulo yakho kwiwebhusayithi namhlanje

	Andivumelani kakhulu (Strongly Disagree)	Andivumelani (Disagree)	Andinangcamango / Ndiphakathi (Neutral)	Ndiyavumelana (Agree)	Ndiyavumelana kakhulu (Strongly Agree)
1. Ndicinga ukuba ndingathanda ukuyisebenzisa rhoqo le nkqubo.	○	○	○	○	○
2. Ndiyifumanise inzima ngokungeyomfuneko le nkqubo.	○	○	○	○	○
3. Ndicinga ukuba bekulula ukuyisebenzisa le nkqubo.	○	○	○	○	○
4. Ndicinga ukuba ndiza kudinga uncedo lomntu onobugcisa ukuze ndikwazi ukuyisebenzisa le nkqubo.	○	○	○	○	○
5. Ndifumanise ukuba imisebenzi eyohlukeneyo kule nkqubo idityaniswe kakuhle.	○	○	○	○	○

Figure 8: Written isiXhosa SUS form used in Phase 2 (page 1).

	Andivumelani kakhulu (Strongly Disagree)	Andivumelani (Disagree)	Andinangcamango / Ndiphakathi (Neutral)	Ndiyavumelana (Agree)	Ndiyavumelana kakhulu (Strongly Agree)
6. Bendicinga ukuba kukho izinto ezininzi ezingahambelaniyo kule nkqubo.	○	○	○	○	○
7. Ndicinga ukuba uninzi lwabantu lungafunda ukuyisebenzisa ngokukhawuleza le nkqubo.	○	○	○	○	○
8. Ndikufumanise kunzima ukuyisebenzisa le nkqubo.	○	○	○	○	○
9. Ndizive ndinokuzithemba okukhulu ukuyisebenzisa le nkqubo.	○	○	○	○	○
10. Kuye kwafuneka ukuba ndifunde izinto ezininzi phambi kokuba ndiqalise ukuyisebenzisa le nkqubo.	○	○	○	○	○

Figure 9: Written isiXhosa SUS form used in Phase 2 (page 2).

E Qualitative Follow-up Questions

Participants providing qualitative feedback were asked questions concerning their interpretation of the questionnaire, perceived differences between the English and translated versions, and language preference. The full question set was:

(1) Can you describe in your own words what this questionnaire is trying to measure?
(2) Did any words or phrases feel unnatural, awkward, or confusing in either version?

(3) Were there any questions where you weren't sure what was being asked?
(4) Do the translated and English versions seem to ask the same thing? Why or why not?
(5) Were there any items where the meaning felt different between the two versions?
(6) Which version felt easier to understand or answer? Why?
(7) Did you answer any items differently between the two versions? What caused that difference?
(8) Which version would you prefer to complete in a real study, and why?
(9) What changes would improve the translated questionnaire?

Not every participant answered every question, and some participants received a shorter core set depending on the mode of follow-up.

F Translation and Adaptation Materials

F.1 Translation Process

The isiXhosa SUS was developed through a multi-stage translation and review process. The stages are summarised in Table 14. The process combined forward translation, linguistic review and synthesis, independent back-translation, discrepancy review, and pretesting before the questionnaire was used in the main study.

Table 14: Stages in the development of the isiXhosa SUS.

Stage	Process
Original English SUS	Original ten-item English SUS used as the source instrument.
Forward translation 1	Initial machine-assisted translation prepared by the researcher using Google Translate.
Forward translation 2	Independent translation produced by a fluent isiXhosa-speaking contributor.
Forward-translation review	Both versions were reviewed for linguistic, cultural, social, and production issues before synthesis.
Synthesis	The two forward versions were compared and combined into a single version by an isiXhosa language specialist.
Back-translation 1	Independent translation of the synthesised isiXhosa version back into English.
Back-translation 2	A second independent English back-translation of the same synthesised version.
Discrepancy review	The back-translations were compared with the original English SUS. Items 3, 5, 6, and 9 were prioritised for further review; Item 2 was also considered in the translator's final decision record.
Pretesting	The synthesised questionnaire was completed by two isiXhosa-speaking pretesters, followed by comprehension questions about the items and response scale.
Follow-up	One pretester completed an additional follow-up concerning interpretation of the five-point response scale and use of the English wording for reference.
Finalisation	The reviewed synthesised wording was retained for the main study.

F.2 Forward-Translation Review

The two forward translations were reviewed before synthesis using an error-coding framework adapted from Pan and Fond [27]. The framework distinguished linguistic problems from cultural, social, and survey-production problems. Table 15 summarises the coding categories used in this review.

Table 15: Categories used to review the forward translations.

Category	Focus of review
Linguistic rules	Lexical choice, morphology, tense, word order, word division, and syntax.
Cultural norms	Whether the source concept or expression functions appropriately within the target-language cultural context.
Social practices	Concepts or practices that may not have a direct equivalent in the target social context.
Production errors	Technical issues such as omissions, typographical errors, or formatting problems introduced during questionnaire preparation.

The review of both forward versions primarily identified linguistic issues. These included changes in tense, unnatural word order, lexical choices, incorrect word division, and omitted wording. Rather than selecting either forward translation in full, the language specialist considered both versions item by item and produced a synthesised translation intended to preserve the meaning of the original SUS while using appropriate isiXhosa wording.

F.3 Back-Translation Discrepancy Review

Two bilingual speakers who had not seen the original English SUS independently back-translated the synthesised isiXhosa version. The resulting English versions were compared with the source instrument. Most items retained the intended overall meaning, but several discrepancies were returned to the language specialist for review. Table 16 summarises the issues recorded in the final decision record.

Table 16: Back-translation discrepancies and translator decisions.

Item	Discrepancy	Translator assessment	Change
2	One back-translation was incomplete, while the other rendered the concept of "complex" in an unsuitable context.	The discrepancies were attributed to the back-translations rather than the synthesised isiXhosa wording.	No
3	One back-translation reversed the item's polarity by rendering "easy" as "hard."	The translator confirmed that the isiXhosa wording communicated "easy" and that the polarity reversal occurred during back-translation.	No
5	"Functions" was rendered as "lessons" in one back-translation and more generically in the other.	The translator judged that the back-translations had used English equivalents that changed the intended context.	No
6	The concept of "inconsistency" was rendered more generally as things not fitting or not working properly.	The translator attributed this difference to the way the back-translators expressed the formal concept in English.	No
9	"Confidence" was rendered as "hope" in one back-translation and "trust" in the other.	The translator judged that both back-translations had altered the intended meaning of the isiXhosa expression for confidence.	No

No changes were made to the synthesised isiXhosa wording following this review. Items 3, 5, 6, and 9 were nevertheless carried forward as priority items for attention during pretesting because they had produced the clearest back-translation discrepancies.

F.4 Worked Translation Example

Item 3 provides a useful example of how discrepancies were handled because one back-translation appeared to reverse the polarity of the original SUS statement.

Table 17: Worked example of the translation and review of SUS Item 3.

Stage	Content
Original English	I thought the system was easy to use.
Forward translation 1	Ndifumanisa ukuba le nkqubo kulula ukuyisebenzisa.
Forward translation 2	Lenkqubo kulula ukuyisebenzisa.
Synthesised isiXhosa	Ndicinga ukuba bekulula ukuyisebenzisa le nkqubo.
Back-translation 1	I think that it was hard to use this programme.
Back-translation 2	I think it was easy to use this app/program.
Discrepancy	Back-translation 1 reversed the intended polarity, rendering "easy" as "hard," while Back-translation 2 preserved the intended positive meaning.
Translator review	The language specialist determined that "easy" had been mistranslated in Back-translation 1 and confirmed that the synthesised isiXhosa statement retained the intended positive meaning.
Final decision	No revision was made. The synthesised wording, *Ndicinga ukuba bekulula ukuyisebenzisa le nkqubo*, was retained.

F.5 Pretesting Procedure

The final synthesised questionnaire was pretested with two isiXhosa-speaking participants before the main study. One pretester evaluated Google Search and the other evaluated an ATM before completing all ten isiXhosa SUS items. The back-translation review had identified Items 3, 5, 6, and 9 as priorities for attention; however, the post-questionnaire prompts were deliberately phrased as general comprehension questions so that participants were not led toward particular problems.

The prompts used after completion of the SUS are shown in Table 18.

Table 18: Comprehension prompts used during initial pretesting.

Focus	Prompt
General comprehension	Were any of the questions difficult to understand? If yes, which ones?
Vocabulary	Were there any words or phrases that were unclear or unfamiliar?
Uncertainty	Was there any question where you were unsure what was being asked?
Ambiguity	Did any question feel confusing or like it could be interpreted in more than one way?
Response scale	Were the response options easy to understand?
Wording changes	Is there anything you would change about the wording of any question?
Cognitive effort	Was there a particular question that you had to think carefully about before deciding on your answer? If yes, which one and why?

Neither pretester reported difficulty understanding the questions, unfamiliar words or phrases, uncertainty about what was being asked, or wording that they wished to change. Both also reported that the response options were easy to understand. However, both indicated that at least some SUS items appeared similar. One pretester commented generally that some questions "sounded the same," while the other specifically identified Items 2 and 8 as similar. These observations concerned perceived similarity between SUS items rather than a specific translation discrepancy.

F.6 Pretesting Follow-up

An additional follow-up was completed by one of the two pretesters to clarify how the five-point agreement scale had been interpreted. The follow-up did not identify a new problem with the meaning of an individual isiXhosa item, but it revealed a possible issue in how the response process had been approached.

The participant reported understanding that the response scale contained five levels of agreement, but described having answered many statements in a more binary "yes/no" or "agree/disagree" manner rather than consistently distinguishing the strength of agreement. The participant suggested that the ranking nature of the response options could be emphasised more explicitly.

The same participant reported understanding the alternation between positively and negatively phrased items. They also reported referring back to the English wording for almost all items, despite finding the isiXhosa questions understandable, in order to confirm their interpretation. They attributed this partly to being more accustomed to completing questionnaires in English. When asked which questions had seemed similar, the participant again identified Items 2 and 8.

The follow-up questions and resulting observations are summarised in Table 19.

Table 19: Summary of the additional pretesting follow-up.

Follow-up focus	Participant response
Understanding of the response scale	Understood the available options but tended to answer as a direct agree/disagree or yes/no decision rather than consistently ranking strength of agreement.
Use of all five categories	Recognised that stronger or weaker responses could apply, but reported that their habitual approach to questionnaires led to more binary choices.
Positive and negative wording	Reported understanding that some items were phrased positively and others negatively.
Reference to English wording	Consulted the English version for almost all items to confirm interpretation, despite reporting that the isiXhosa items themselves were understandable.
Similar items	Again identified Items 2 and 8 as sounding similar.
Suggested instruction	Recommended emphasising that respondents should use the scale to indicate degree of agreement rather than approaching each item as a yes/no question.

Because this additional follow-up involved only one pretester, these observations are treated as an individual pretesting case rather than evidence of a general response pattern.

G Recruitment Materials

G.1 Recruitment Poster

Figure 10 reproduces the recruitment poster used during participant recruitment.

G.2 WhatsApp Recruitment Invitation

The substantive wording of the short-form invitation circulated through WhatsApp and other informal recruitment networks is reproduced below. Live invitation links and contact details have been omitted:

> **Hi there! Are you a speaker of English, isiXhosa, or Afrikaans?**
> We're running a UCT research study where participants evaluate the usability of a few different apps or devices in their own language, as well as a second language (for multilingual participants), in order to study the effects that language may have on the result.
> **We're looking for people who:**
> - Are 18 years or older.
> - Fall into one of the following groups:



Figure 10: Recruitment poster used for the SUSEVAL study.

(1) English speakers at any level of proficiency, from basic to fluent;
(2) isiXhosa native speakers (first-language or home-language); or
(3) Afrikaans native speakers (first-language or home-language).

What you will do:

- Evaluate six different apps or devices using a short survey.
- Participate in a short interview afterwards about your experience.

The entire process should take about 35 minutes, split into two 15-minute surveys and a short 5-minute interview afterwards. Everything will happen online via WhatsApp and Microsoft Forms, so you can complete the study in your own time.

If you are interested, you can join our WhatsApp community here: *[WhatsApp community link omitted from the appendix].*

Once you join, please complete the consent form found in the community description. A researcher will contact you with further details soon after.

Feel free to email us if you have any questions.

G.3 Formal Invitation to Participate

The substantive wording of the longer invitation used for formal recruitment and circulation through university and community contacts is reproduced below. Contact details and live links have been omitted. This invitation relates to the broader SUSEVAL team project and therefore includes the English and Afrikaans study arms in addition to the isiXhosa component reported in this paper:

Invitation to Participate
SusEval: Evaluating the System Usability Scale for South African Languages

Dear isiXhosa, Afrikaans, and English speakers,

We are Inhle Cele, Dhiren Chetty, and Yazeed Adams, Honours students in the Department of Computer Science at the University of Cape Town. For our research, we are developing and evaluating the System Usability Scale (SUS), a widely used tool for measuring how usable a system feels, for isiXhosa and Afrikaans, and testing the English-language SUS with non-native speakers.

Translated versions of the scale are not always checked for reliability, and it is not yet clear how sensitive the scale is when used across different languages. This project aims to help close that gap.

Who can participate?

- 18 years or older;
- First-language speaker of isiXhosa or Afrikaans and speaker of English, **or** first- or second-language speaker of English;
- Open to all levels of English proficiency, from non-native/basic to fluent speakers;
- UCT student, staff member, or member of the public;
- Access to an active WhatsApp account; and
- Internet access for WhatsApp and online forms.

Participation details

We are looking to recruit 25 participants per language group. Sessions will be conducted remotely via WhatsApp, allowing you to participate at a scheduled time that suits you best. The entire session will take approximately 35 minutes and consists of two phases:

- **Phase 1 (15 mins):** You will complete a short system task followed by the standard written English SUS questionnaire accessed via Microsoft Forms (no sign-in required).
- **Phase 2 (15 mins):** Depending on your group selection, you will complete a second task followed by either a translated written SUS (Afrikaans or isiXhosa) or an orally administered English SUS.
- **Wrap-up (5 mins):** A brief set of qualitative questions about your experience and thoughts on the scale.

To join, participants were directed to the SUSEVAL WhatsApp Community.

Accessibility

If you have specific communication needs or accessibility requirements, please let us know when you get in touch, and we will do our best to accommodate you.

Please note:

- This project was approved by the Faculty of Science Research Ethics Committee (SCI/02801/2026).
- The interview may be audio recorded and transcribed where applicable.
- Participants were not compensated for participation.

- Participants could contact the student researchers or project supervisor with any questions.

We look forward to working with you!
Inhle Cele, Dhiren Chetty & Yazeed Adams
Honours Students
Department of Computer Science, University of Cape Town

H Consent and Demographic Materials

Participants completed a consent and demographic form before beginning the study. Equivalent English and isiXhosa versions were made available through Microsoft Forms. The original online forms were visually spread across multiple pages; the substantive content is condensed below for readability.

H.1 Consent Information

The consent form informed participants of the following:

- the purpose of the study and the nature of participation;
- that participation involved evaluating digital systems and completing usability questionnaires;
- that participation was voluntary and that participants could withdraw before anonymisation without penalty;
- that identifying information would be kept separate from research responses and that findings would be reported anonymously;
- that study communication would take place primarily through WhatsApp and Microsoft Forms;
- that WhatsApp and Microsoft Forms are third-party platforms and therefore carry some residual privacy risk;
- that participants could receive feedback on the study findings;
- that audio recording applied only where an oral interview or oral study component was undertaken; and
- that consent to participate and, where applicable, consent to audio recording were collected explicitly.

Participants were also asked to provide their name, contact number, consent selection, and digital signature. These identifying details were stored separately from the analysis data.

H.2 English Demographic Questions

The English demographic form contained the following questions:

(1) What is your age?
(2) What is your gender?
(3) What is your current occupation/status?
(4) What is your highest completed level of education?
(5) Which province do you currently live in?
(6) What is your home language—the language you learned first, growing up?
(7) Which other languages do you speak, read, or write, even if not fluently?
(8) How would you rate your proficiency in isiXhosa, Afrikaans, and English?
(9) If English is not your first language, at what age did you start learning it?
(10) Which language would you feel most comfortable completing this study in?
(11) Do you have regular access to a smartphone with WhatsApp?
(12) How often do you use WhatsApp?
(13) How would you describe your overall comfort using apps or websites?

H.3 isiXhosa Demographic Questions

The equivalent isiXhosa demographic form included:

(1) Uneminyaka emingaphi?
(2) Sithini isini sakho?
(3) Yintoni umsebenzi/isimo sakho ngoku?
(4) Leliphi inqanaba eliphezulu lemfundo oliqhubileyo?
(5) Uhlala kweliphi iphondo ngoku?
(6) Luluphi ulwimi lwakho lwasekhaya—ulwimi owalufunda kuqala?
(7) Zeziphi ezinye iilwimi ozithethayo, ozifundayo, okanye ozibhalayo?
(8) Ungazichaza njani izinga lakho lolwazi kwezi lwimi zilandelayo: isiXhosa, Afrikaans, nesiNgesi?
(9) Ukuba isiNgesi asilolwimi lwakho lokuqala, wawuneminyaka emingaphi xa waqala ukusifunda?
(10) Loluphi ulwimi ongaziva ukhululekile ukusisebenzisa ekugqibezeleni olu phononongo?
(11) Ngaba unokufikelela ngokurhoqo kwifowuni ekrelekrele ene-WhatsApp?
(12) Uyisebenzisa kangaka i-WhatsApp?
(13) Ungakuchaza njani ukukhululeka kwakho jikelele xa usebenzisa ii-app okanye amawebhusayithi?

Response categories. Age, gender, occupation, education, language proficiency, most comfortable language, WhatsApp use, and technology comfort were collected using predefined categorical response options, with "Other" or free-text fields where applicable. Home language permitted multiple selections.

For educational attainment, the principal categories used in analysis were High School, Undergraduate degree/diploma, and Postgraduate degree. The most-comfortable-language question offered isiXhosa, Afrikaans, and English. Language proficiency was collected using ordered categories ranging from no proficiency/basic proficiency through conversational and fluent proficiency to first/native language.

Administration. The consent and demographic questions were administered through Microsoft Forms. The material above condenses the substantive content of the original forms for readability rather than reproducing the interface layout.

AI Usage Statement

Google Translate was used by the author to prepare the first forward isiXhosa version of the SUS. This machine-assisted version was not adopted directly; it was compared with an independently produced

forward translation, synthesised by a qualified isiXhosa language specialist, and subsequently reviewed through back-translation and pretesting.

ChatGPT and Claude were used as research-support tools during data cleaning, analysis, and writing. Their assistance included helping organise and check cleaned data, reviewing R code and statistical interpretation, supporting the organisation and refinement of qualitative codes and themes, identifying methodological inconsistencies or repetition, and drafting or revising wording for sections of the paper. AI-generated suggestions were checked against the study records, analysis outputs, participant material, and cited sources before inclusion. The author made the final methodological and interpretive decisions and remains responsible for the study design, data handling, analysis, citations, conclusions, and final submission.